



Professional Data Analysis & Intelligent Reporting

# Professional Data Analysis Report

Comprehensive Analysis & Insights

|                   |                                   |
|-------------------|-----------------------------------|
| Report Generated: | December 23, 2025                 |
| Platform:         | Insightify Professional Analytics |
| Engine Version:   | 3.0 (Enhanced AI Analytics)       |
| Report Status:    | Confidential                      |

Generated by Insightify™ - AI-Powered Data Analysis Platform  
Transforming Data into Actionable Insights

## ■ Executive Summary

### Report Overview:

This comprehensive data analysis report presents key insights and actionable metrics from the provided dataset. Our analysis covers **50,000** records across **11** columns, utilizing advanced statistical methods and visualization techniques.

### Dataset Composition:

- Numeric Variables: **5** columns
- Categorical Variables: **6** columns
- Temporal Variables: **0** columns
- Data Completeness: **0** missing values detected
- Data Uniqueness: **0** duplicate rows identified
- Memory Footprint: **19.85 MB**

## ■ Key Insights & Findings

### Our analysis has uncovered the following significant patterns:

1. Transmission is highly concentrated: 'Manual' represents 50.3% of data
2. Sales\_Classification is highly concentrated: 'Low' represents 69.5% of data

## ■ Numeric Analysis

| Column        | Mean      | Median    | Std Dev  | Min      | Max       | IQR       |
|---------------|-----------|-----------|----------|----------|-----------|-----------|
| Year          | 2017.02   | 2017.00   | 4.32     | 2010.00  | 2024.00   | 8.00      |
| Engine_Size_L | 3.25      | 3.20      | 1.01     | 1.50     | 5.00      | 1.70      |
| Mileage_KM    | 100307.20 | 100388.50 | 57941.51 | 3.00     | 199996.00 | 100452.25 |
| Price_USD     | 75034.60  | 75011.50  | 25998.25 | 30000.00 | 119998.00 | 45193.50  |
| Sales_Volume  | 5067.51   | 5087.00   | 2856.77  | 100.00   | 9999.00   | 4949.25   |

## ■ Categorical Analysis

### Category: Model

Unique Values: 11 | Most Frequent: '7 Series' (9.3% of data)

| Value    | Count | Percentage |
|----------|-------|------------|
| 7 Series | 4,666 | 10.2%      |
| i3       | 4,618 | 10.1%      |
| i8       | 4,606 | 10.1%      |
| 3 Series | 4,595 | 10.1%      |
| 5 Series | 4,592 | 10.1%      |
| X1       | 4,570 | 10.0%      |
| X3       | 4,497 | 9.9%       |
| X5       | 4,487 | 9.8%       |
| M5       | 4,478 | 9.8%       |
| X6       | 4,478 | 9.8%       |

### Category: Region

Unique Values: 6 | Most Frequent: 'Asia' (16.9% of data)

| Value         | Count | Percentage |
|---------------|-------|------------|
| Asia          | 8,454 | 16.9%      |
| Middle East   | 8,373 | 16.7%      |
| North America | 8,335 | 16.7%      |
| Europe        | 8,334 | 16.7%      |
| Africa        | 8,253 | 16.5%      |
| South America | 8,251 | 16.5%      |

### Category: Color

Unique Values: 6 | Most Frequent: 'Red' (16.9% of data)

| Value  | Count | Percentage |
|--------|-------|------------|
| Red    | 8,463 | 16.9%      |
| Silver | 8,350 | 16.7%      |
| Grey   | 8,348 | 16.7%      |
| White  | 8,304 | 16.6%      |
| Black  | 8,273 | 16.5%      |
| Blue   | 8,262 | 16.5%      |

### Category: Fuel\_Type

Unique Values: 4 | Most Frequent: 'Hybrid' (25.4% of data)

| Value | Count | Percentage |
|-------|-------|------------|
|-------|-------|------------|

|          |        |       |
|----------|--------|-------|
| Hybrid   | 12,716 | 25.4% |
| Petrol   | 12,550 | 25.1% |
| Electric | 12,471 | 24.9% |
| Diesel   | 12,263 | 24.5% |

Category: Transmission

Unique Values: 2 | Most Frequent: 'Manual' (50.3% of data)

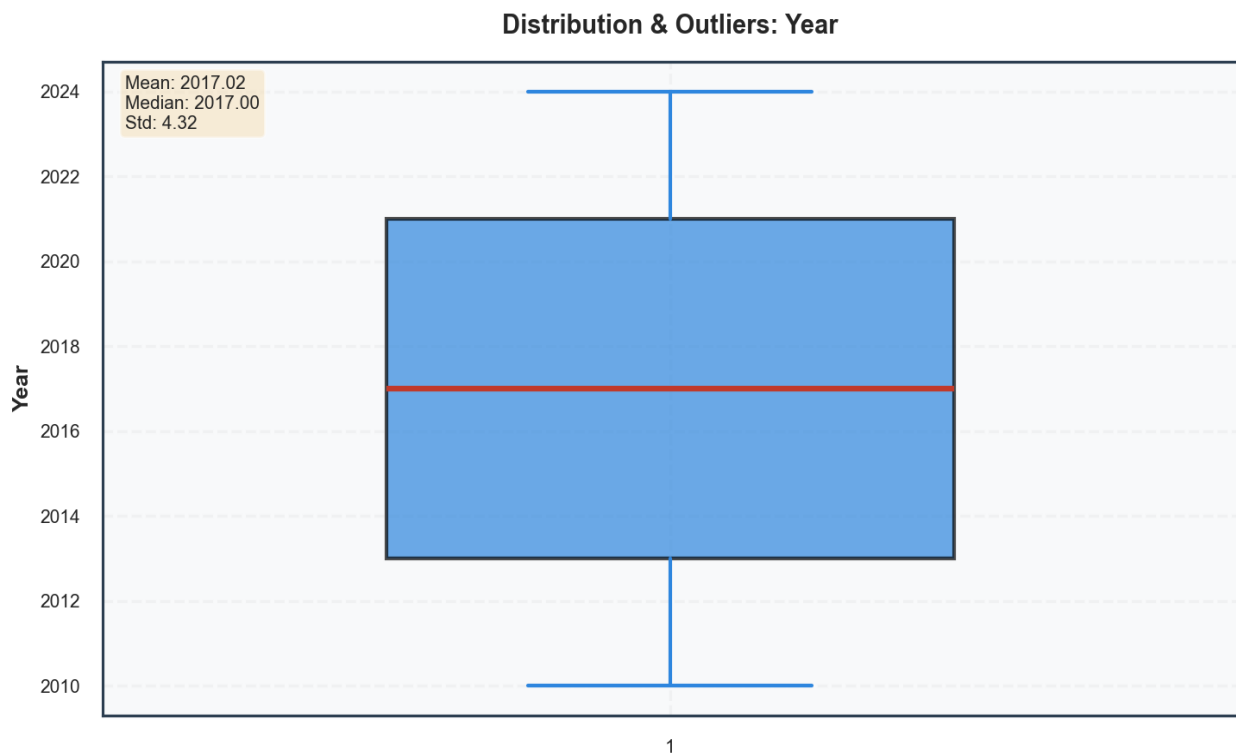
| Value     | Count  | Percentage |
|-----------|--------|------------|
| Manual    | 25,154 | 50.3%      |
| Automatic | 24,846 | 49.7%      |

## ■ Correlation Analysis

*No significant correlations detected in the dataset.*

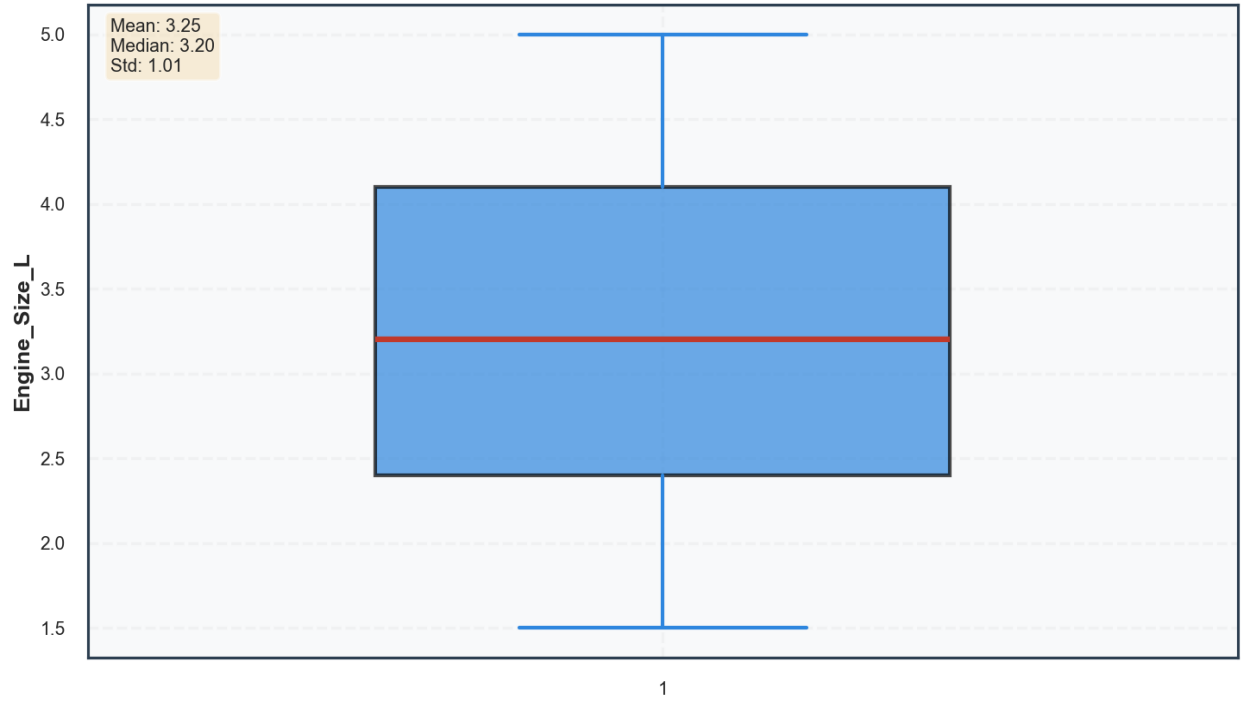
## ■ Data Visualizations

### Boxplot Year

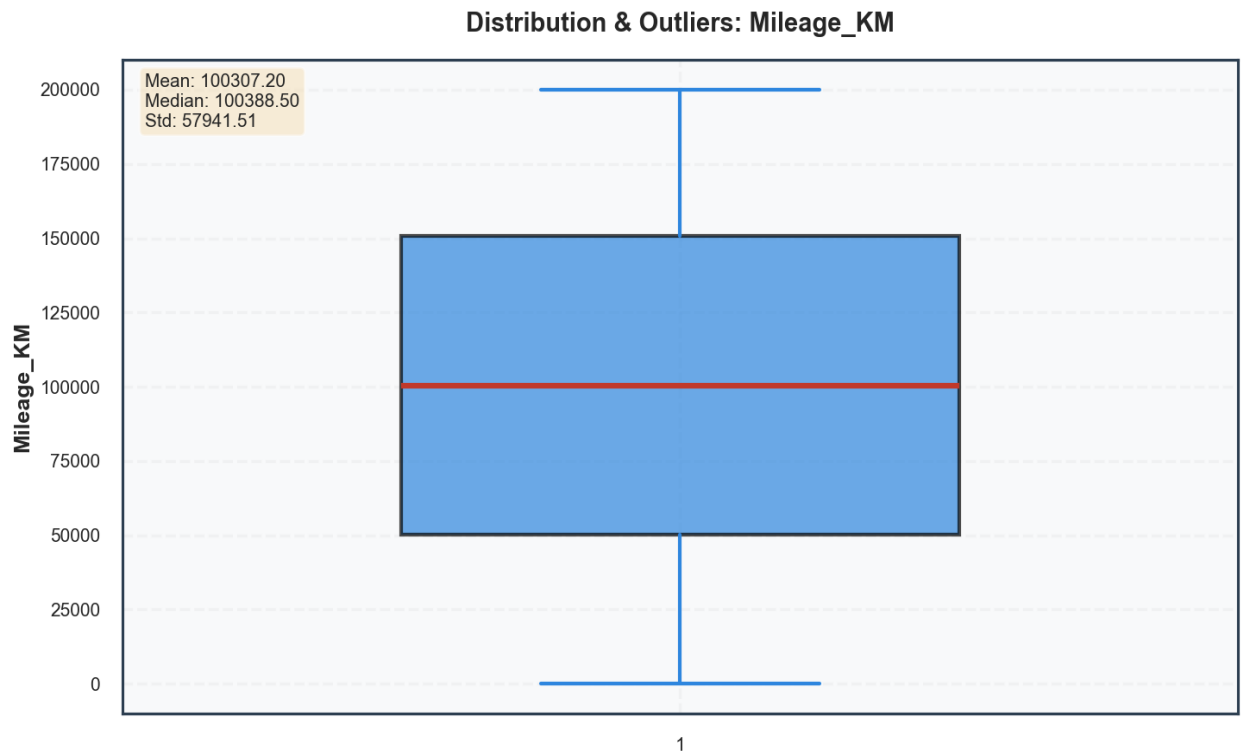


### Boxplot Engine Size L

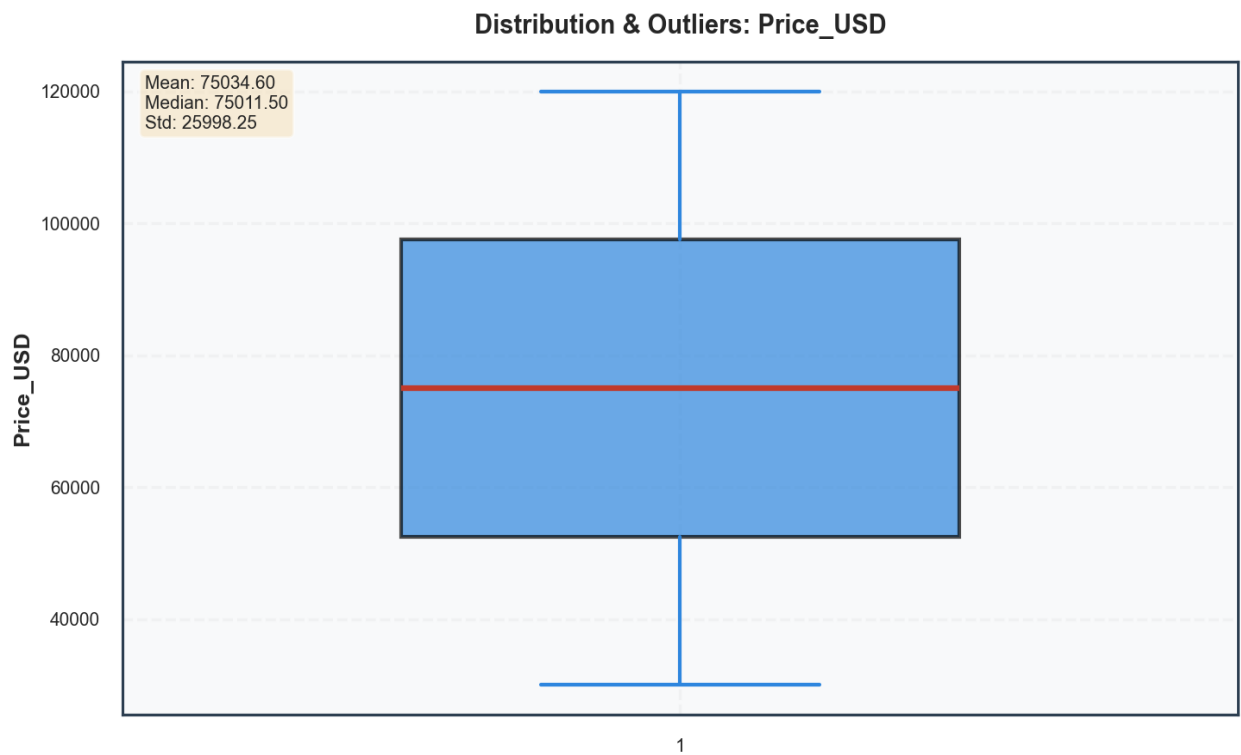
Distribution & Outliers: Engine\_Size\_L



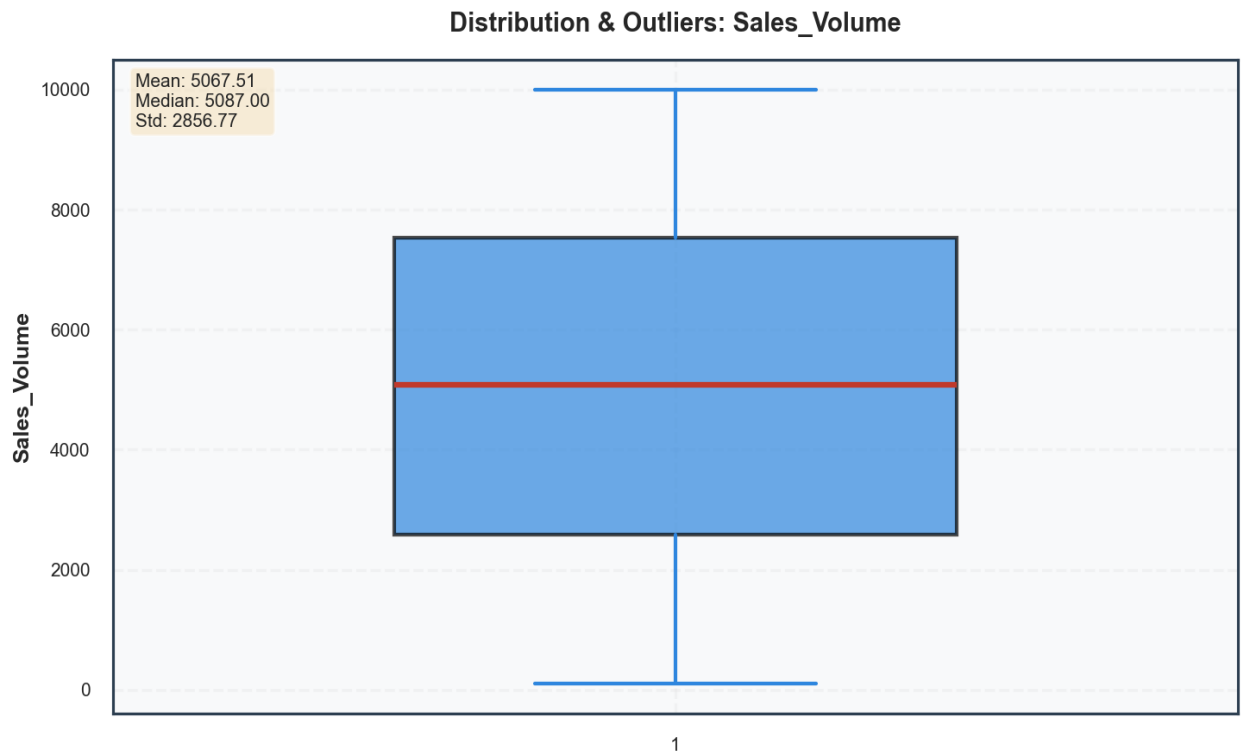
## Boxplot Mileage\_Km



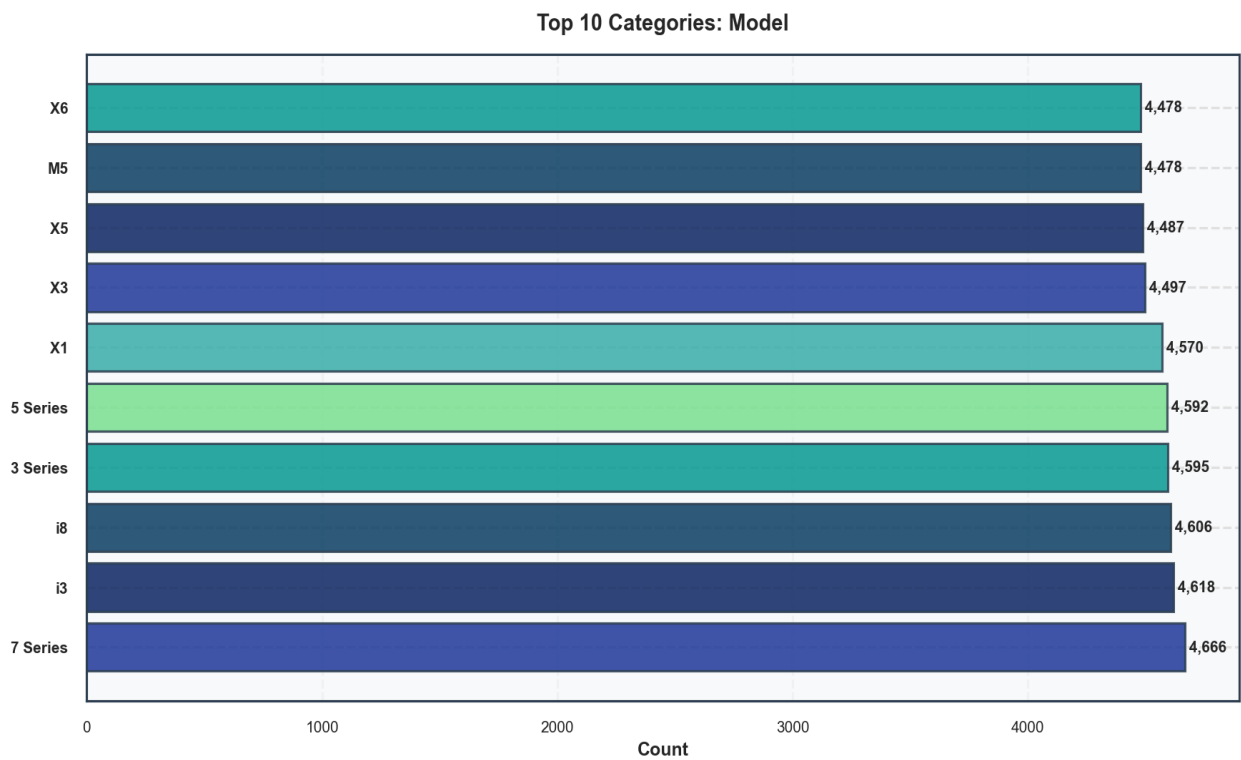
## Boxplot Price\_Usd



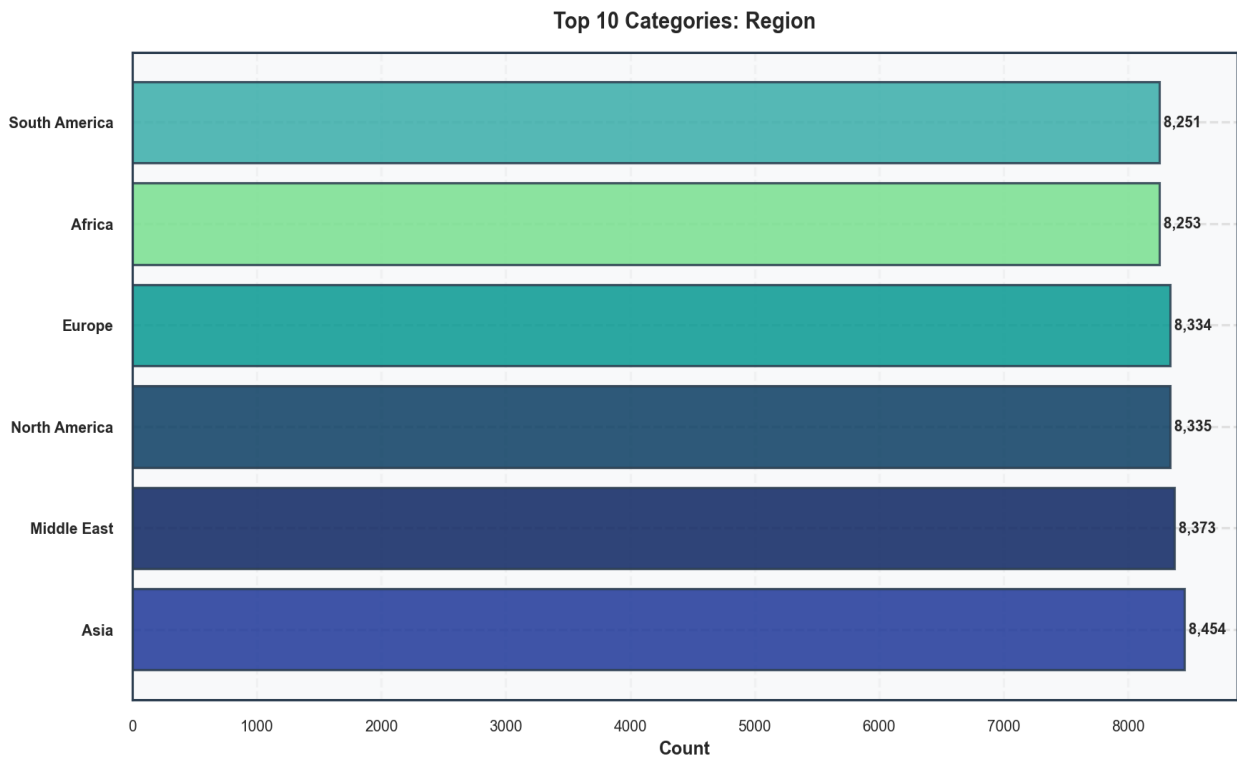
## Boxplot Sales Volume



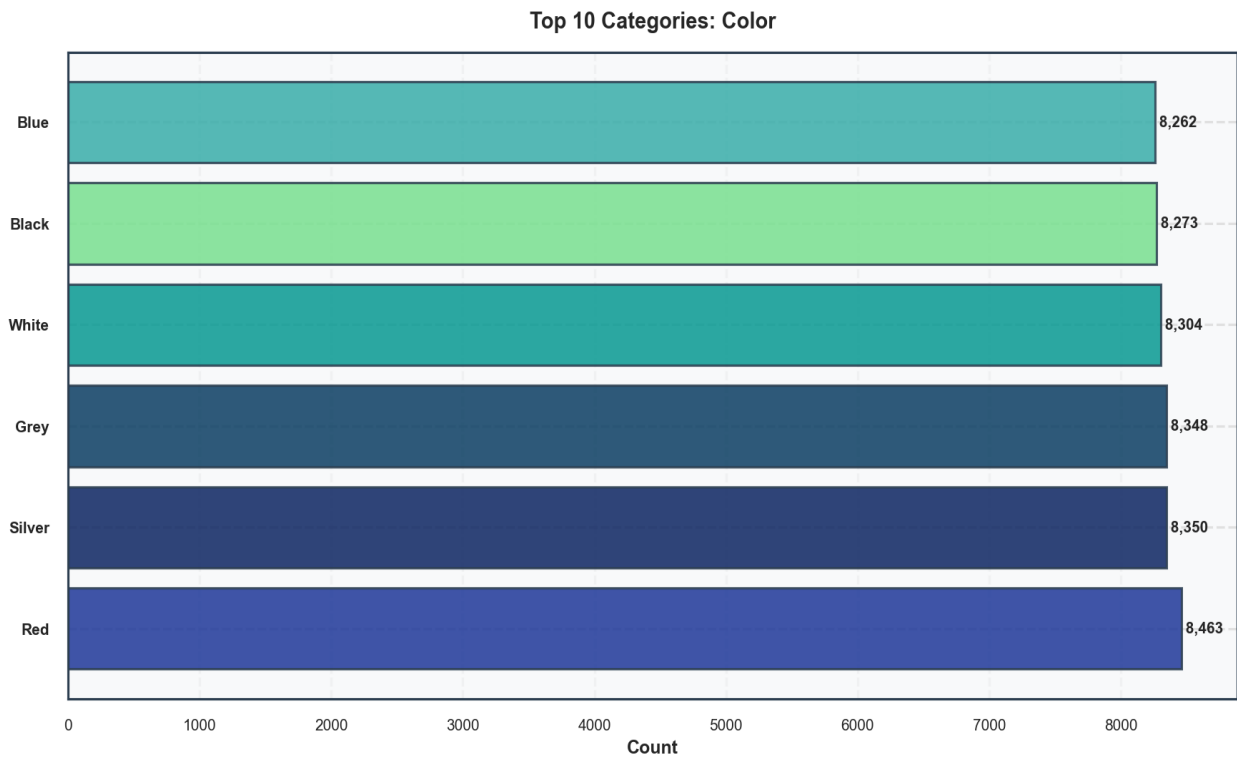
## Bar Model



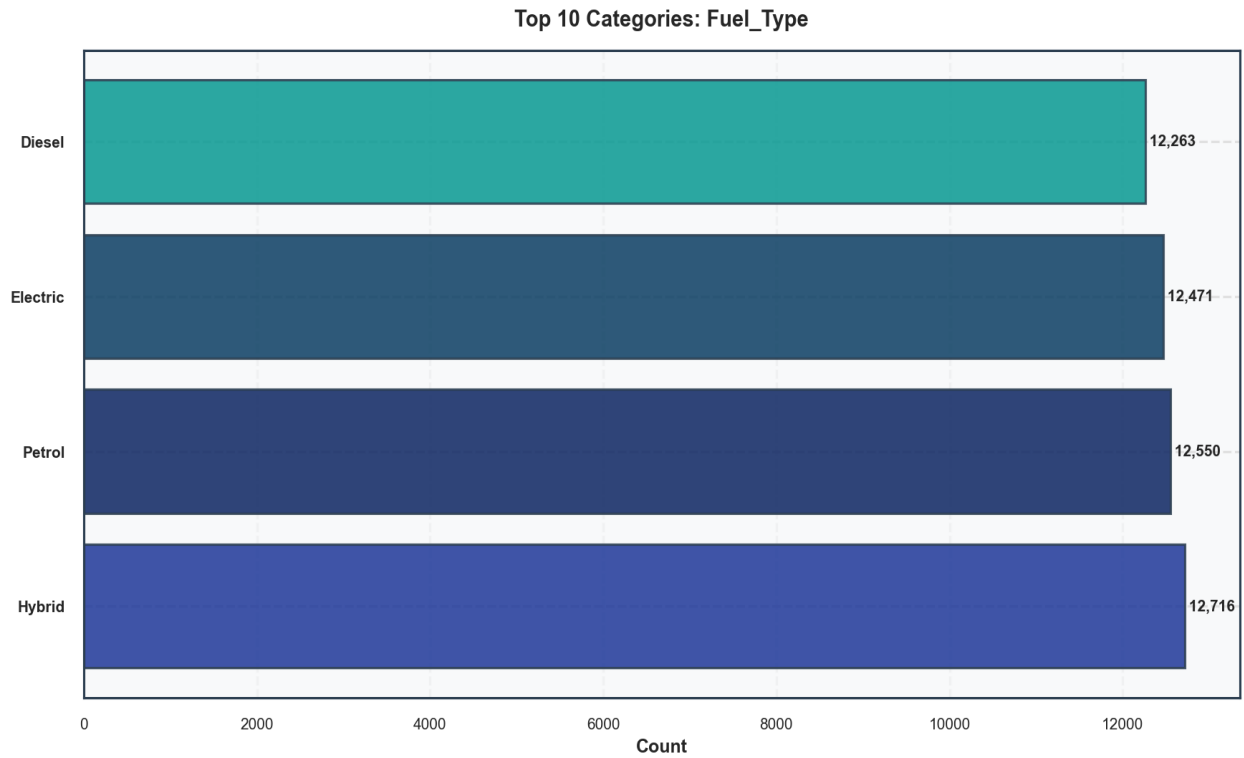
Bar Region



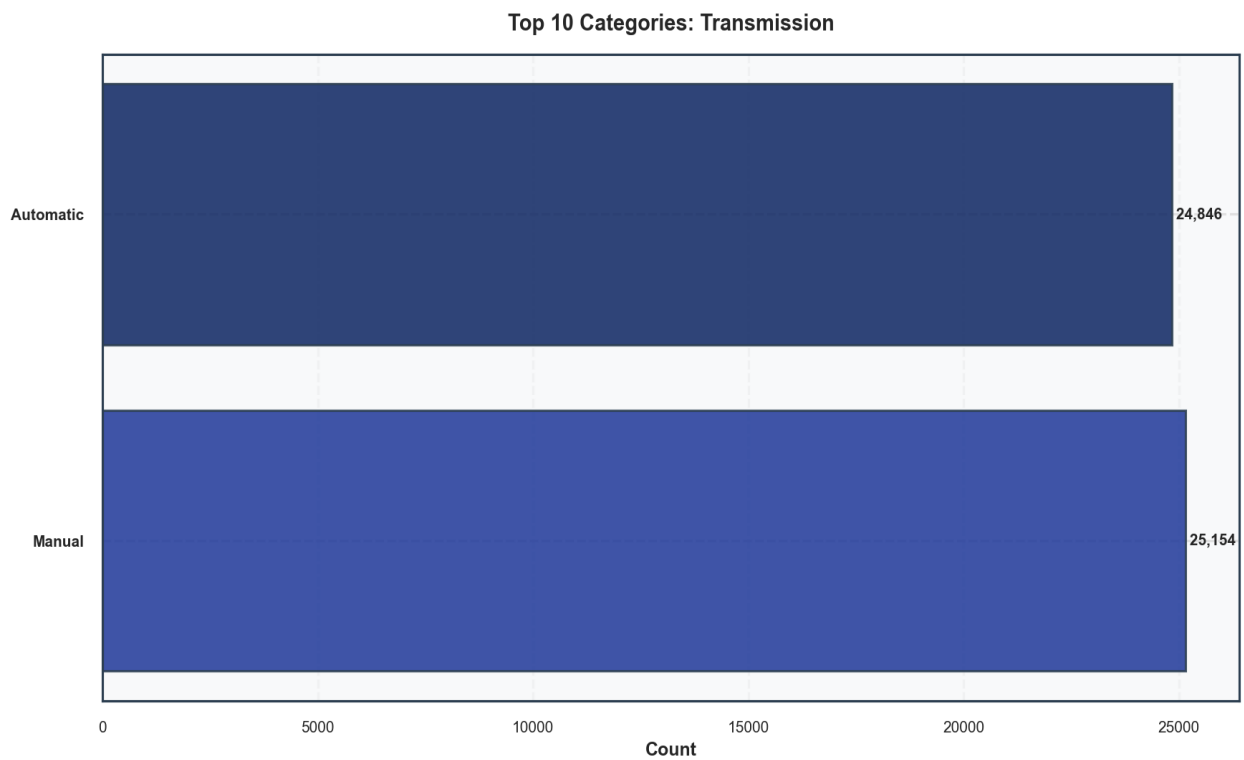
Bar Color



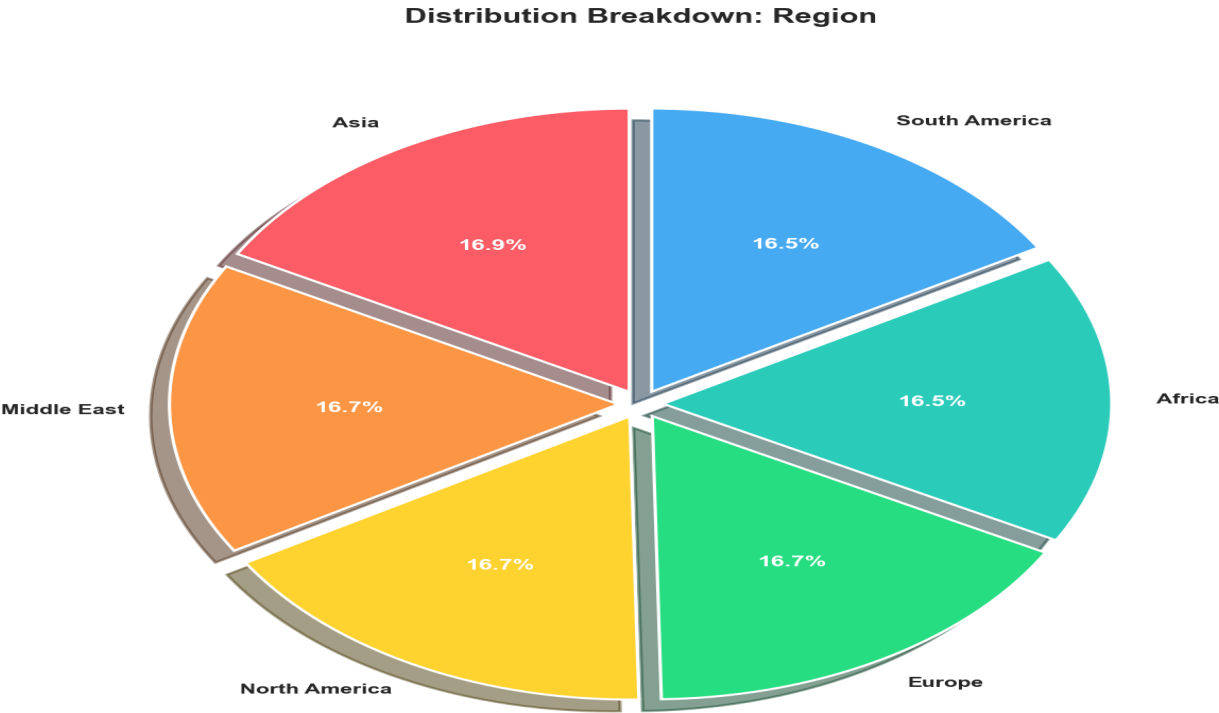
## Bar Fuel Type



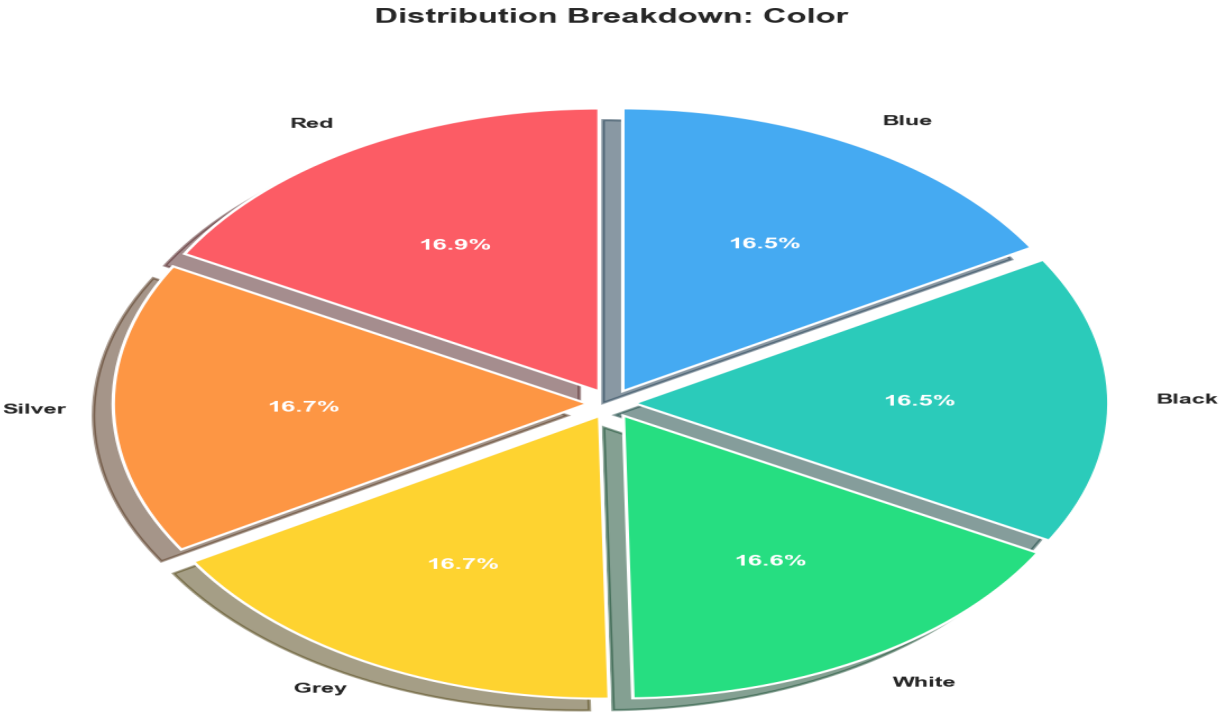
## Bar Transmission



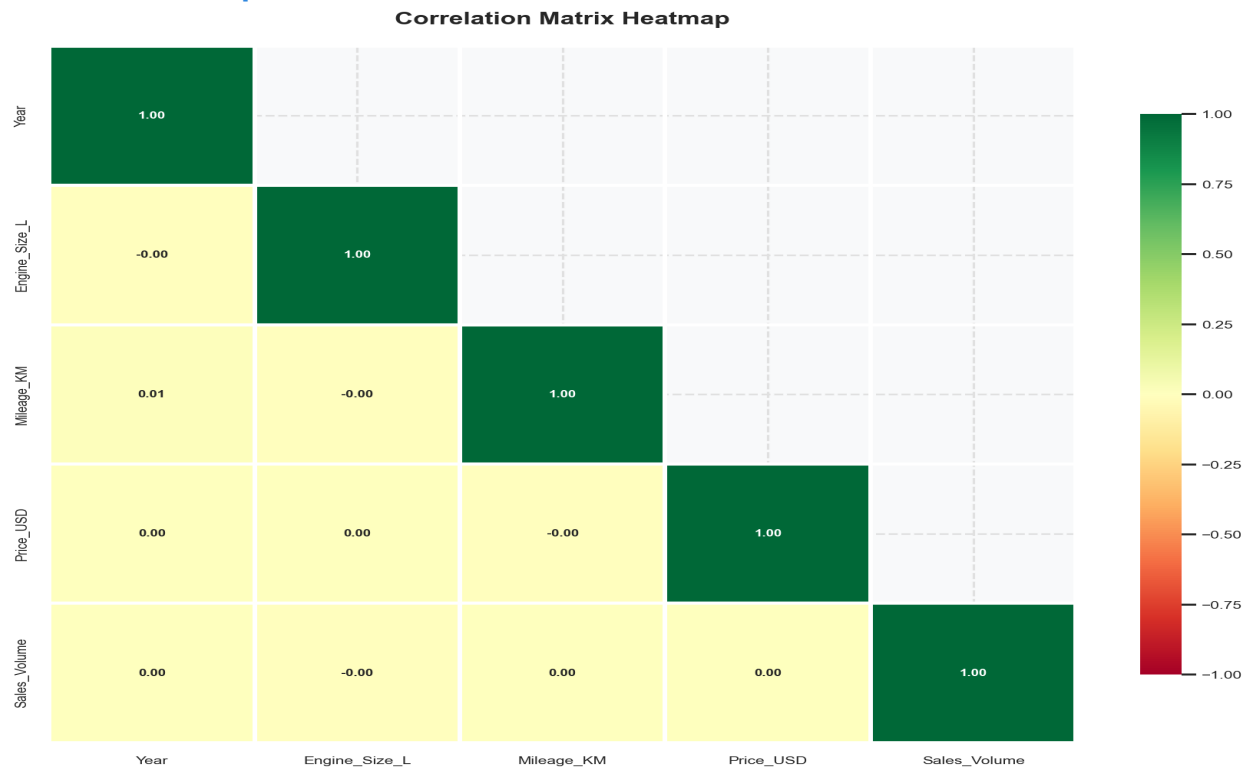
Pie Region



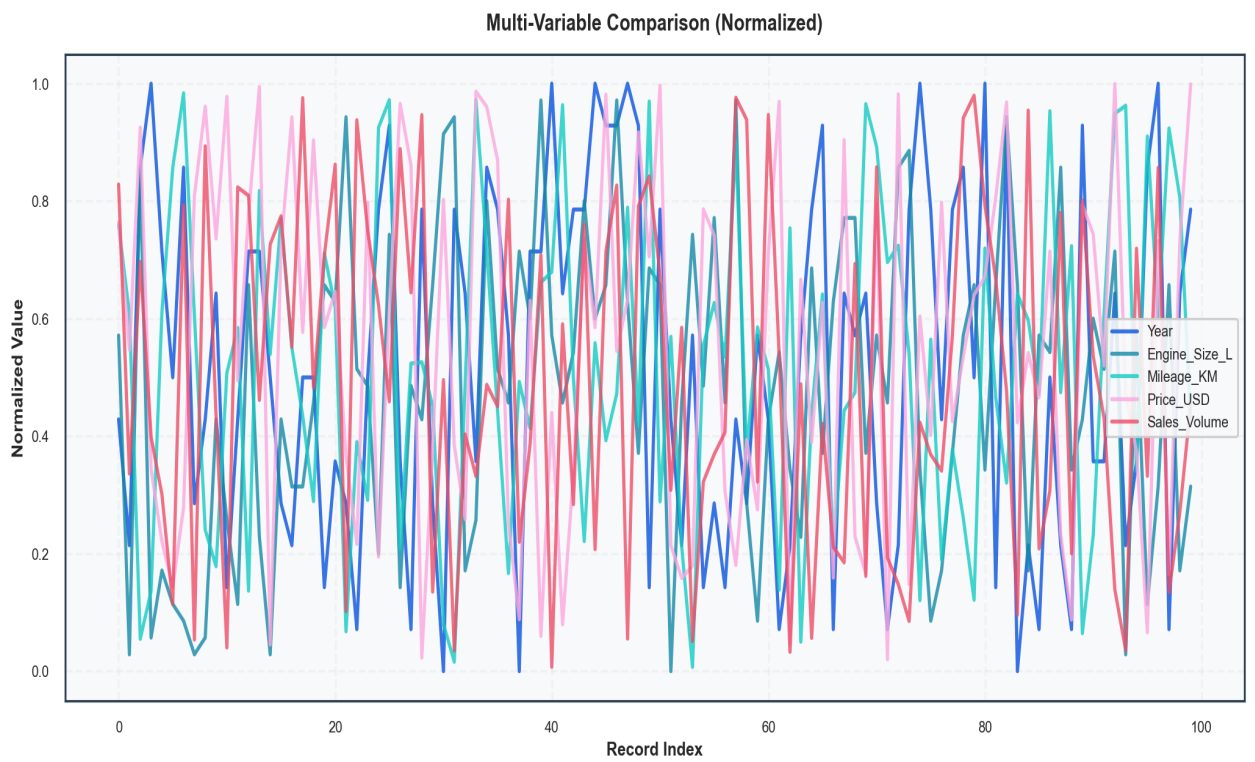
Pie Color



Correlation Heatmap



Multiline Comparison



# ■ Conclusions & Recommendations

## Key Findings:

### 1. Data Quality Assessment:

The dataset has undergone comprehensive quality checks. Review the Executive Summary for critical insights regarding data completeness, duplicate records, and overall integrity. Address any missing values or duplicates identified before proceeding with predictive modeling.

### 2. Statistical Distribution Analysis:

Our numeric analysis reveals the central tendencies, variability, and distribution patterns across all quantitative variables. Box plots highlight potential outliers that may require further investigation or treatment depending on your analytical objectives.

### 3. Correlation & Relationship Discovery:

The correlation analysis has identified statistically significant relationships between variables. Strong positive or negative correlations ( $|r| > 0.7$ ) indicate potential predictive relationships that could be leveraged in machine learning models or business intelligence applications.

## Actionable Recommendations:

✓ **Investigate High-Correlation Pairs:** Variables with strong correlations warrant deeper analysis to determine causality versus mere association. Consider domain expertise when interpreting these relationships.

✓ **Address Data Quality Issues:** Implement data cleaning protocols for missing values and duplicates. Consider imputation strategies or removal based on the nature and extent of missing data.

✓ **Leverage Visualizations for Stakeholder Communication:** The generated charts provide clear, professional representations of your data suitable for executive presentations and decision-making forums.

✓ **Monitor Categorical Distributions:** Highly concentrated categories or imbalanced classes may require special handling in predictive models (e.g., SMOTE for oversampling, stratified sampling).

✓ **Temporal Pattern Exploration:** If time-series trends are present, consider seasonal decomposition and forecasting techniques to predict future patterns.

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